Arbitrarily Long Gaps Between Prime Powers

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Abstract

For each integer $L \geq 1$, there are infinitely many runs of L consecutive positive integers, none equal to a prime power. In fact their starting positions contain an explicit arithmetic progression of positive natural density. Consequently the gaps between consecutive terms of OEIS A000961 are unbounded. The proof is a direct application of the Chinese remainder theorem.

OEIS A000961 consists of 1 and the integers p^k with p prime and $k \geq 1$. The exceptional term 1 is irrelevant to gaps sufficiently far out.

Theorem 1 (Prime-power-free blocks)**.** *For each $L \geq 1$, there exist positive integers N_0 and M such that for every integer $t \geq 0$ and every $1 \leq i \leq L$, the integer*

$$N_0 + tM + i$$

has at least two distinct prime divisors. In particular none of these integers belongs to A000961.

Proof. Choose $2L$ pairwise distinct primes

$$q_1, r_1, q_2, r_2, \ldots, q_L, r_L,$$

and set $m_i = q_i r_i$ and $M = \prod_{i=1}^{L} m_i$. Because the moduli $m_1, \ldots, m_L$ are pairwise coprime, the Chinese remainder theorem supplies a residue class $N_0 \pmod M$ satisfying the simultaneous congruences

$$N_0 \equiv -i \pmod{m_i}, \qquad 1 \leq i \leq L. \tag{1}$$

Choose a positive representative N_0. Every $N = N_0 + tM$ satisfies the same congruences. Therefore

$$q_i \mid (N + i) \quad \text{and} \quad r_i \mid (N + i)$$

for each i. These are distinct prime divisors. A positive integer divisible by two different primes is not a prime power p^k; moreover $N + i > 1$. Hence all L consecutive integers $N + 1, \ldots, N + L$ are missing from A000961. □

Corollary 1 (Unbounded consecutive-term gaps)**.** *If $1 = a_1 < a_2 < a_3 < \cdots$ enumerates A000961, then*

$$\sup_{j \geq 1} (a_{j+1} - a_j) = \infty.$$

Proof. Given L, take a block in Theorem 1 with N large enough that there is a prime power at or below N. There is one after $N + L$ because there are infinitely many primes. The consecutive A000961 terms bracketing the block are at most N and at least $N + L + 1$, respectively, so their difference is at least $L + 1$. Since L is arbitrary the gaps are unbounded. □

Corollary 2 (Positive density for each fixed block length)**.** *For every fixed L, the set of positive integers N such that $N + 1, \ldots, N + L$ are not prime powers has lower natural density at least $1/M > 0$, where M is the product in the construction.*

Proof. The set contains every sufficiently large $N \equiv N_0 \pmod M$, which has natural density exactly $1/M$. □

Example. Taking $L = 2$ and $m_1 = 6$, $m_2 = 35$ yields $N_0 = 173$: the consecutive numbers $174 = 2{\cdot}3{\cdot}29$ and $175 = 5^2 \cdot 7$ are not prime powers. The next examples are obtained by repeating this residue class modulo 210.

Scope, provenance and verification

This is an elementary, known type of Chinese-remainder construction; it makes no claim to settle the separate question, raised in the prime-power literature, of whether three *consecutive prime powers* in arithmetic progression occur infinitely often. The assertion proved concerns *blocks avoiding* prime powers, not arithmetic progressions *of* prime powers. *AI authorship notice.* This document was prepared by an AI system. The displayed proofs were algebraically reviewed and finite examples checked computationally, but there has been no independent human peer review. Numerical checks do not substitute for a proof. No research-priority or originality claim is made.

References

1. OEIS Foundation, A000961: Powers of primes, including the convention that 1 is included.
2. OEIS Foundation, A250267: Three consecutive prime powers in arithmetic progression (distinct open-ended motivation).
3. L. Panaitopol, "Some of the properties of the sequence of powers of prime numbers," *Rocky Mountain J. Math.* **31** (2001), 1407–1415; publication link.